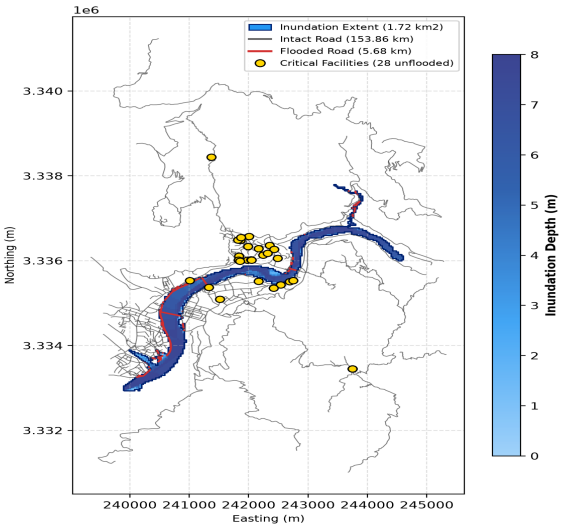
PravahX Dam Breach & Inundation Brief

Rishikesh Ganga Reach (Uttarakhand, India) | Tier-0 HAND Model

Run ID: tier0_ganga_accepted
Date: 2026-10-25 17:52 UTC
Status: ACCEPTED SCREENING

HYPOTHETICAL SCENARIO - TIER-0 SCREENING
Ganga Reach near Rishikesh (UTM 44N / EPSG:32644) | Peak Flow: 5,000 m3/s



DATA PROVENANCE & ENGINE:
- **DEM:** Copernicus GLO-30 (30 m STAC AWS)
- **CRS:** EPSG:32644 (UTM 44N) | **AOI:** 5x5 km
- **Hydraulic Forcing:** Peak Flow 5,000.0 m3/s, Manning n=0.035
- **Population:** WorldPop 2020 100m Constrained (BSGM IND)
- **Infrastructure:** OpenStreetMap (Snapshot: 2026-10-22, ODbL)

1. Hydraulic Screening Envelope

Inundation Area	**1.72 km2 (171.9 ha)**
Peak Inflow Rate	**5,000.0 m3/s**
Maximum / Mean Depth	**7.73 m / 6.31 m**
Median (P10 - P90)	**7.23 m** (3.12 - 7.73 m)
Arrival Time	None (Static Tier-0)

2. Population & Asset Exposure

Exposed Population	**2,696 people (9.8%)**
• Depth 0.0-0.5 m [UNVERIFIED]	230 (8.5%)
• Depth 0.5-1.5 m [UNVERIFIED]	253 (9.4%)
• Depth > 1.5 m [UNVERIFIED]	2,213 (82.1%)
Flooded Roads (depth > 0.3m)	**17.34 km of 159.5 km (10.9%)**
Flooded Buildings	**5** of 93 (OSM coverage incomplete)
Critical Facilities	**0 of 28 (depth = 0.00 m)**
Monetary Loss Valuation	**NOT RUN** (No verified depth-damage curve)

3. Network Evacuation Assessment

Vehicle Impassable Cutoff	Depth > 0.30 m [UNVERIFIED]
Severed Road Edges	**100** of 914 (14.8 km)
Reachable Building Origins	**2 of 5 (Routes: 34.4 m, 4.7 m)**
Unreachable Building Origins	**3 of 5 (Gorge road severed)**

MODEL LIMITATIONS & UNCERTAINTIES (MANDATORY DISCLAIMER):

*1. **Static Inundation Envelope:** Tier-0 HAND operates on static equilibrium and does not resolve hydrodynamic wave propagation, backwater effects, or arrival times.*
*2. **Synthetic Rating Curve:** Inundation depth is derived from normal-depth Manning hydraulics along a uniform reach without bathymetric cross-section survey calibration.*
*3. **DEM Water-Surface Artefact:** Depths are measured above the DEM water surface; true depth in the channel is greater (DEM records water surface, not bed). Copernicus GLO-30 reflects water surface elevation during acquisition; dry-season channel bed conveyance is underestimated.*
*4. **Operational Scope:** Generated strictly for preliminary disaster risk screening and multi-agency table-top planning. NOT for structural engineering or design use.*